



Semester 1 Examinations 2016 / 2017

Exam Code(s) 4BCT1, MECS1, CSD1, SPE1
Exam(s) B.Sc. Degree (Computer Science and Information Technology)
M.Sc. in Computer Science (Data Analytics)

Module Code(s) CT561
Module(s) Systems Modelling and Simulation

Paper No. I

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Instructions:

Answer any 3 questions. All questions carry equal marks.

Duration 2hrs
No. of Pages 5 (Including Cover Page)
Department(s) Information Technology

Requirements Graph Paper

1. (a) Define negative feedback, and show how the stock management structure (with a single stock) can be viewed as a negative feedback system. (No equations need to be defined, but the anchor and the adjustment components should be clearly identified).

[8]

- (b) Construct a stock and flow model from the following description of student workload (there is no requirement to list equations to the model).

- The *assignment backlog* is increased by *new assignments* and reduced by *completions*.
- *Additional rework* also increases the assignment backlog.
- As the *assignment backlog* increases, so too does the *work pressure*
- There are two student responses to increasing work pressure:
 - The *time per assignment* is reduced
 - The *workweek* is increased
- As the time per assignment increases:
 - The *completions* reduce
 - The *Additional rework* reduces
- As the workweek increases:
 - *Completions* increase
 - *Fatigue* increases
- As *fatigue* increases, so too does *time per assignment*.

[10]

- (c) Based on the stock and flow model in part(b):

- Show the feedback loops and calculate their polarity, by tracing an increase in a variable all the way around a feedback loop, and observing its direction of change.
- Discuss how the feedback loops can help identify the implications of leaving assignments to the last possible minute, rather than working on them in a consistent way throughout a semester.

[7]

2. (a) What is the difference between a stock and a flow, and how are the two concepts related? Identify stocks and associated flows that can be used to model programmers in a software company.

[5]

- (b) Consider the derivative $dy/dt = 2t + 10$

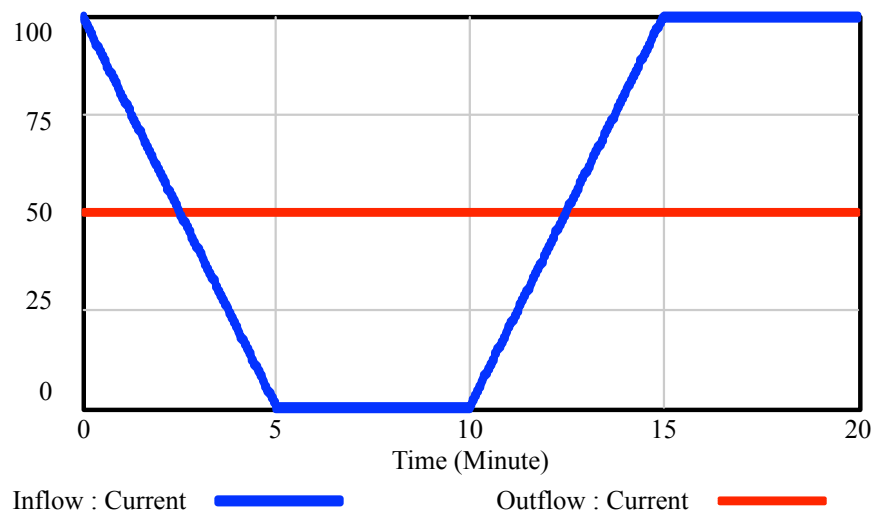
Show this as a stock and flow model, with equations, assuming the initial value of the stock is 0.

Integrate the derivative over the interval $[0,5]$ using Euler's method, with $DT = 0.50$.

[10]

- (c) Based on the flow variables shown in the diagram (the outflow is constant at 50), and assuming that the initial value of the stock is 100, perform the following:

- Calculate, and show, the net flow
- Use graphical integration to plot the stock over time, clearly showing the behaviour mode for each segment (e.g. *increasing at an increasing rate, etc.*)



[10]

3. (a) Suppose we have a town with 100,000 ($=N$) individuals, of which 1% were infectious with measles, with $R_0 = 18$ and recovery delay (D) = 7 days. Calculate the:

- Effective per capital contact rate β
- Force of infection λ
- Herd Immunity Threshold

[5]

- (b) Design a stock and flow model (with equations) to simulate the spread of influenza. Assume that the effective contact rate is 2, and that the average recovery delay is 2 days. The model should have the following features:

- Its core structure should be based on the *Susceptible – Infected – Recovered* model.
- It should contain a vaccination feature, whereby healthcare staff can administer vaccines on a daily basis. Important variables include the *number of healthcare workers*, and the *average vaccination rate per health worker per day*. Vaccines (a consumable resource) can only be administered if there are health workers, and if there is an available supply of vaccines.
- Healthcare workers can become ill, and so may be unavailable for work to administer vaccines. On average, 10% of staff fall ill each week, and they recover after an average delay of 3 days.

[15]

- (c) Show how the stock and flow model would need to be modified in order to represent different age cohorts, specifically:

- 0-14
- 15-44
- 45-64
- 65+

[5]

4. (a) Describe the general approach to formulating effects in a system dynamics model for a dependent variable Y and a set of independent variables $(X_1, X_2, \dots, X_N)$, where all variables also have a reference value.

[5]

- (b) Construct a stock and flow model (with feedbacks) for a University, with two stocks: Students and Staff. The relevant flows are:

- Staff are increased by hires, and reduced by retirements.
- Students are increased by enrolments, and reduced by attrition, and by graduations.

An important performance indicator variable is *student to staff ratio*. This impacts the hiring rate decision, and also informs student enrolment.

An additional features of this variable is:

1. It has a *reference value* of 20 (i.e. 20 students for every academic staff member.)
2. It has been shown to impact student attrition levels. When the student to staff ratio is 20, the *reference attrition rate* is 0.02.
3. If the staff to student ratio goes as high as 30, then the student attrition rate rises to 5 times its reference value.
4. If the staff student ratio drops to 15, then the student attrition rate will drop to half its reference value.

Based on the information, in [1-4], construct an appropriate set of equations (including your assumptions) to model the *effect of the student staff ratio on the student attrition rate*.

[15]

- (c) Comment on how the feedback loops identified in part(b) can inform decision making by the University Management Team.

[5]